The systems approach & control

ENGINEERING WITH MR KELLY

Higher · Engineering contexts & challenges · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ draw a universal system diagram (input, process, output) for a given system
- ☐ draw sub-system diagrams, including feedback paths
- ☐ describe the difference between open loop and closed loop control
- ☐ draw and describe a closed loop control diagram: comparator, controller, driver, output, sensor
- ☐ describe proportional and two-state (on/off) control, and when each suits a situation

I DO — Worked example

Read how a full-mark answer is built

An automatic hand dryer switches on when hands are detected under it and switches off after 10 seconds.

1 Draw a universal system diagram for the hand dryer. (3 marks)

Full-mark answer each ✓ is one mark

Input: presence of hands ✓ → Process: control/timing ✓ → Output: heat and moving air ✓.

Why marks are not awarded

"Input: IR sensor → Process: microcontroller → Output: heater and fan."

0 marks: these are components, not system-level inputs and outputs.

WE DO — Guided practice

Use the starters and word bank to help you

The hand dryer runs for 10 seconds every time it is triggered.

2 State whether the dryer uses open loop or closed loop control, and justify your answer. (2 marks)

Sentence starters

1. …
2. …

Word bank

Feedback · whether

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) On a universal system diagram, the middle block is labelled...
☐ Control ☐ Process ☐ Driver ☐ Sensor

b) In a closed-loop heater, which component measures the actual temperature?
☐ Heater ☐ Relay ☐ Temperature sensor ☐ Comparator

c) At the comparator, the error signal is...
☐ Set + feedback ☐ Set − feedback ☐ Feedback × gain ☐ Output ÷ input

d) Which control type gives the smoothest approach to the set point?
☐ Open loop ☐ Two-state ☐ Proportional ☐ None

4 Key terms: write the letter of the matching description next to each term.

1. **Universal system diagram**	____	**A** One block of a system, e.g. sensor, controller, driver, output device.
2. **Sub-system**	____	**B** A sensor feeds back the actual output; the controller compares it with the set level and adjusts.
3. **Open loop control**	____	**C** Input → process → output.
4. **Closed loop control**	____	**D** Set value minus feedback value, found at the comparator.
5. **Feedback**	____	**E** The signal from the output returned to the input/controller for comparison.
6. **Error signal**	____	**F** No feedback: the output is not checked.

An incubator must keep eggs at 37.5 °C ± 0.2 °C.

5 Explain why two-state (on/off) control is unsuitable for the incubator. (2 marks)

A motor speed controller uses proportional control.

6 Describe how proportional control works. (2 marks)

A greenhouse heater keeps the temperature at 22 °C using a thermistor.

7 Describe how the closed loop control system keeps the temperature at 22 °C. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?